GRADE 1 · BLOCK CODING

Build the Code

Write the blocks to reach the target

Name: .... Date: ....

★ **I can write code that lands Bit on a target number.**

You write the code

Now YOU write the code. The start block is given. Fill each empty block with a move — forward 1, forward 2, back 1, or back 2 — so Bit lands on the target number. Then run your code to check.

Bit's number path

0	1	2	3	4	5

Build 1 — start at 0, reach 3

start at 0

[ ]

[ ]

Write a move in each empty block.

1 Reach 3

Fill the 2 empty blocks so Bit goes from 0 to 3. Then run your code. Which number does it land on? (It should be 3.)

Build 2 — start at 5, reach 2

start at 5

This time Bit needs to go back.

2 Predict, then reach 2

Fill the 2 empty blocks so Bit goes from 5 to 2. PREDICT the landing number from your blocks, then run to check.

Build 3 — start at 1, reach 5

start at 1

3 You try — challenge

Fill the 2 empty blocks so Bit goes from 1 to 5. Run it to check. Can you find a DIFFERENT pair of blocks that also reaches 5? Write it on the line.

Big idea

To write code, you pick the blocks and put them in order to reach your goal. Then you run it to test it. Sometimes more than one code reaches the same number.

TEACHER GUIDE · GRADE 1 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Students write their own short instructions — choosing and ordering move blocks — to make Bit reach a target number, run it to test, then find a second set of blocks that reaches the same target. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 with arrow/number cards so students can lay out and walk their blocks before writing them.

How to lead it (about 20 minutes)

1. Show them first (you do). For Build 1, count from 0 to 3 — that is 3 steps forward. Show one way to split it: "forward 2, then forward 1." Run it to land on 3.
2. Do one together (we do). Exercise 1: students fill the 2 blocks to reach 3. Exercise 2: start at 5, reach 2 (3 steps back) — predict, then run.
3. Let them try alone (you do). Exercise 3: start at 1, reach 5 (4 steps forward); then find a second set of blocks that also reaches 5.
4. Big idea: "You wrote instructions to reach a goal, then ran them to test — and there can be more than one correct answer."

Answer key (these are the verified correct answers)

Exercise 1 — start 0 → 3: any two forward blocks that add to 3, e.g. forward 2, forward 1.

Exercise 2 — start 5 → 2: any two back blocks that add to 3, e.g. back 2, back 1.

Exercise 3 — start 1 → 5: e.g. forward 2, forward 2; a different way: forward 1, forward 3. Both reach 5.

Why it works: the start sets Bit's number and the moves run in order; the forward/back amounts must add up to the distance from start to target.

If students get stuck — and ways to adjust

Watch for: moves that overshoot or fall short. Have students count the steps from start to target first, then split that into 2 blocks.

Make it easier: allow one move block (e.g. forward 3) instead of two.

Make it harder: ask for a 3-block answer, or one that uses both a forward and a back block.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student writes blocks that run to the target and can offer a second set that also reaches 5.